**PlanBench-XL** Model and gradient-update time per seed

| Method | Qwen2.5-7B | | Llama-3.1-8B | | Qwen3.8-27B | |
|---|---|---|---|---|---|---|
| | Model (s) | Gradient (s) | Model (s) | Gradient (s) | Model (s) | Gradient (s) |
| Frozen Base | 0 | 0 | 0 | 0 | 0 | 0 |
| AREW / T3 | 82.7 | 2.58 | 113.6 | 2.76 | 322.4 | 2.46 |
| GiGPO | 83.2 | 2.56 | 112.5 | 2.70 | 320.5 | 2.48 |
| InfoPO | 82.9 | 3.29 | 109.5 | 3.47 | 317.0 | 3.15 |
| WorldEvolver | 75.0 | 0 | 99.5 | 0 | 289.3 | 0 |
| PoS / EnvProbe | 0 | 0 | 0 | 0 | 0 | 0 |
| Step-DPO | 85.1 | 4.35 | 115.7 | 4.63 | 337.4 | 4.24 |
| Outcome-PG | 84.4 | 3.29 | 112.9 | 3.55 | 318.2 | 3.22 |
| Outcome-Value | 83.9 | 2.72 | 113.2 | 2.88 | 317.8 | 2.54 |
| Outcome-RLOO | 84.2 | 3.28 | 112.9 | 3.51 | 318.3 | 3.12 |
| Info-RL Neighbor | 83.9 | 3.41 | 111.2 | 3.64 | 318.2 | 3.10 |
| Unstratified | 83.0 | 2.30 | 113.4 | 2.23 | 322.0 | 2.07 |
| Unweighted | 86.3 | 2.21 | 117.2 | 2.20 | 347.1 | 2.18 |
| **SPAR** | **87.1** | **2.24** | **118.6** | **2.36** | **346.5** | **2.17** |